$3d^8$

September 1, 2026

1 $\left|3d_{xz}^2 3d_{xy}^2 3d_{x^2-y^2}^2 3d_{yz}^{\alpha} 3d_{z^2}^{\alpha}\right\rangle$

Here are the CSFs:

$$|1,1,1\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|2,1,1\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^{\alpha} 3d_{z^2}^2 3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|3,1,1\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^{\alpha} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^2\right\rangle$$

$$|4,1,1\rangle = \left|3d_{xz}^2 3d_{yz}^{\alpha} 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|5,1,1\rangle = \left|3d_{xz}^2 3d_{yz}^{\alpha} 3d_{xy}^2 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^2\right\rangle$$

$$|6,1,1\rangle = \left|3d_{xz}^2 3d_{yz}^{\alpha} 3d_{xy}^{\alpha} 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|7,1,1\rangle = \left|3d_{xz}^{\alpha} 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|8,1,1\rangle = \left|3d_{xz}^{\alpha} 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^2\right\rangle$$

$$|9,1,1\rangle = \left|3d_{xz}^{\alpha} 3d_{yz}^2 3d_{xy}^{\alpha} 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|10,1,1\rangle = \left|3d_{xz}^{\alpha} 3d_{yz}^{\alpha} 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|11,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|12,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^{\alpha} 3d_{z^2}^2 3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^{\beta} 3d_{z^2}^2 3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|13,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|14,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|15,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|16,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$|17,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|18,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|19,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$|20,1,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$|21,1,-1\rangle = \left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|22,1,-1\rangle = \left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|23,1,-1\rangle = \left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle$$

$$|24,1,-1\rangle = \left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|25,1,-1\rangle = \left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle$$

$$|26,1,-1\rangle = \left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$|27,1,-1\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|28,1,-1\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle$$

$$|29,1,-1\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$|30,1,-1\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$|31,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|32,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|33,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2\right\rangle$$

$$|34,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|35,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2\right\rangle$$

$$|36,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|37,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|38,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2\right\rangle$$

$$|39,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|40,0,0\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|41,0,0\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0\right\rangle$$

$$|42,0,0\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2\right\rangle$$

$$|43,0,0\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|44,0,0\rangle = \left|3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|45,0,0\rangle = \left|3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

The Ground State CSFs are 5 15 25

$\langle 5,1,1\,|L_x|\,1,1,1\rangle = I$

Δg_{xx}

$+\zeta\Delta_{5-1}^{-1}$

$\langle 5,1,1\,|L_y|\,3,1,1\rangle = -I$

$\langle 5,1,1\,|L_y|\,10,1,1\rangle = \sqrt{3}I$

Δg_{yy}

$+\zeta\Delta_{5-3}^{-1}$
$+3\zeta\Delta_{5-10}^{-1}$

$\langle 5,1,1\,|L_z|\,8,1,1\rangle = I$

Δg_{zz}

$+\zeta\Delta_{5-8}^{-1}$

For a multiplicity of 3 we have the following:

$D = \langle 1,1|1,1\rangle - \langle 1,0|1,0\rangle$

$E = \langle 1,-1|1,1\rangle$

$$\begin{aligned}\left\langle 1,1\left|\hat{H}_{eff}\right|1,1\right\rangle =& -\zeta^2\,(\\ &+\frac{1}{4}\Delta_{8}^{-1}+\frac{1}{8}\Delta_{11}^{-1}+\frac{1}{8}\Delta_{13}^{-1}+\frac{3}{8}\Delta_{20}^{-1}+\frac{1}{8}\Delta_{31}^{-1}+\frac{1}{8}\Delta_{33}^{-1} \\ &+\frac{3}{8}\Delta_{40}^{-1}+\frac{3}{4}\Delta_{42}^{-1}+\frac{3}{4}\Delta_{44}^{-1}\;)\end{aligned}$$

$$\begin{aligned}\left\langle 1,0\left|\hat{H}_{eff}\right|1,0\right\rangle =& -\zeta^2\,(\\ &+\frac{1}{8}\Delta_{1}^{-1}+\frac{1}{8}\Delta_{3}^{-1}+\frac{3}{8}\Delta_{10}^{-1}+\frac{1}{8}\Delta_{21}^{-1}+\frac{1}{8}\Delta_{23}^{-1}+\frac{3}{8}\Delta_{30}^{-1} \\ &+\frac{1}{4}\Delta_{38}^{-1}\;)\end{aligned}$$

$$\begin{aligned}\left\langle 1,-1\left|\hat{H}_{eff}\right|1,1\right\rangle =& -\zeta^2\,(\\ &+\frac{1}{8}\Delta_{11}^{-1}-\frac{1}{8}\Delta_{13}^{-1}-\frac{3}{8}\Delta_{20}^{-1}-\frac{1}{8}\Delta_{31}^{-1}+\frac{1}{8}\Delta_{33}^{-1} \\ &+\frac{3}{8}\Delta_{40}^{-1}-\frac{3}{4}\Delta_{42}^{-1}-\frac{3}{4}\Delta_{44}^{-1}\;)\end{aligned}$$

$$
\begin{aligned}
D =\zeta^2 \, (& \\
& + \frac{1}{8}\Delta_{1}^{-1} + \frac{1}{8}\Delta_{3}^{-1} - \frac{1}{4}\Delta_{8}^{-1} + \frac{3}{8}\Delta_{10}^{-1} - \frac{1}{8}\Delta_{11}^{-1} - \frac{1}{8}\Delta_{13}^{-1} \\
& - \frac{3}{8}\Delta_{20}^{-1} + \frac{1}{8}\Delta_{21}^{-1} + \frac{1}{8}\Delta_{23}^{-1} + \frac{3}{8}\Delta_{30}^{-1} - \frac{1}{8}\Delta_{31}^{-1} - \frac{1}{8}\Delta_{33}^{-1} \\
& + \frac{1}{4}\Delta_{38}^{-1} - \frac{3}{8}\Delta_{40}^{-1} - \frac{3}{4}\Delta_{42}^{-1} - \frac{3}{4}\Delta_{44}^{-1} \,)
\end{aligned}
$$

$$
\begin{aligned}
E =\zeta^2 \, (& \\
& - \frac{1}{8}\Delta_{11}^{-1} + \frac{1}{8}\Delta_{13}^{-1} + \frac{3}{8}\Delta_{20}^{-1} + \frac{1}{8}\Delta_{31}^{-1} - \frac{1}{8}\Delta_{33}^{-1} \\
& - \frac{3}{8}\Delta_{40}^{-1} + \frac{3}{4}\Delta_{42}^{-1} + \frac{3}{4}\Delta_{44}^{-1} \,)
\end{aligned}
$$